

# 3.02 MWp Utility-Scale Solar PV Power Plant

Comprehensive Stakeholder DPR Addendum & Detailed Engineering Architecture

Location: Santuri, Purulia District, West Bengal | Off-taker: WBSEDCL | Interconnection: 33 kV Dedicated Feeder

DC INSTALLED  
ARRAY

**3.02 MWp**

AC COMPACT  
STATION

**4.3 MVA  
Skid**

TOTAL CAPEX  
INVESTMENT

**₹10.80 Cr**

BANK DEBT /  
TENURE

**₹7.57 Cr (10  
Yrs)**

### 1.1 Photovoltaic Cell Physics & Semiconductor Dynamics

The plant utilizes **Waaree Energies 700W N-Type TOPCon (Tunnel Oxide Passivated Contact) Bifacial Modules**. N-type TOPCon technology replaces traditional P-type PERC architectures to eliminate Light-Induced Degradation (LID) and Light and elevated Temperature Induced Degradation (LeTID).

|                                                  |                                |
|--------------------------------------------------|--------------------------------|
| Front Glass & Anti-Reflective Coating            |                                |
| N-Type Silicon Substrate (Base)                  |                                |
| Ultra-Thin Tunnel Oxide Layer (~1.5 nm SiO2)     | <-- Low Resistance Passivation |
| Highly Doped n+ Polysilicon Layer (Rear Contact) | <-- Prevents Recombination     |
| Rear Dual-Glass Encapsulation                    |                                |

- Passivation Physics:** TOPCon integrates an ultra-thin tunnel oxide layer (~1.5 nm) combined with a thin film of heavily doped n+ polysilicon on the rear surface. This tunnel oxide allows majority carriers (electrons) to tunnel through while passivating surface defects, reducing carrier recombination velocity to < 10 cm/s.
- Temperature Coefficient:** The temperature coefficient of maximum power ( $\gamma$ ) is -0.30%/°C, outperforming PERC (-0.35%/°C). This preserves generation efficiency under Santuri’s peak summer ambient temperatures of up to 42°C (where module operating temperatures reach 62°C).
- Bifaciality & Albedo Capture:** The modules exhibit a bifaciality factor of  $80 \pm 5\%$ . The rear side captures diffuse horizontal irradiance and ground-reflected albedo. Given Purulia's terrain (dry gravelly soil with an albedo coefficient of ~0.20), rear-side irradiance adds a continuous 8% to 10% current boost ( $I_{mp}$  increases from 16.36 A to ~17.80 A under peak conditions), elevating effective string output without increasing thermal voltage stress.

### 1.2 String Configuration & Voltage Safety Limits

The DC array consists of **4,340 total procured modules** (140 pallets of 31 modules each), deployed as **4,312 active modules** across **154 parallel strings of 28 modules in series**, with **28 cold spares** retained in the site depot.

```

[ 28 Modules in Series ]
(Voc = 51.20V STC -> 53.12V @ 10°C)
+---+ +---+ +---+ +---+
| M |->| M |->...->| M |->| M |---+
+---+ +---+ +---+ +---+ |
|
[ 154 Parallel Strings Total ] +--> [ Underground DC Bus ]
| (Max Voc: 1,487.36V DC)
+---+ +---+ +---+ +---+ | (Operating Vmp: 1,198V DC)
| M |->| M |->...->| M |->| M |---+
+---+ +---+ +---+ +---+

```

String Voltage Sizing Calculations

- **STC Open-Circuit Voltage (Voc at 25°C):** 51.20 V DC
- **Cold Winter Extreme Temperature (10°C in Purulia):** Temperature differential  $\Delta T = 10^{\circ}\text{C} - 25^{\circ}\text{C} = -15^{\circ}\text{C}$ .
- **Voltage Temperature Coefficient ( $\beta_{\text{Voc}}$ ):**  $-0.25\%/^{\circ}\text{C}$  ( $-0.128\text{ V}/^{\circ}\text{C}$ ).
- **Maximum Cold Voc per Module:**  $\text{Voc}(10^{\circ}\text{C}) = 51.20\text{ V} + (-15 \times -0.128\text{ V}) = 53.12\text{ V DC}$
- **Maximum Cold String Voltage at 10°C:**  $\text{V}_{\text{string,max}} = 28\text{ modules} \times 53.12\text{ V} = \mathbf{1,487.36\text{ V DC}}$

**Voltage Safety Verdict:** This configuration enforces a strict safety margin below the **1,500 V DC** insulation ceiling of IEC 61730 Class A standards, DC solar cables, and inverter MPPT input limits, eliminating over-voltage insulation breakdown risks during cold winter morning startups.

1.3 Inverter Architecture & Closed-Loop Thermal Dynamics

The project utilizes the **FIMER PVS980-CS 4.3 MVA Compact Station**, housing a single high-power **PVS980-58 central inverter core**, a step-up transformer, and medium-voltage switchgear inside a 40-foot High Cube containerized steel enclosure.

```

+-----+
|                                     |
|               FIMER PVS980-CS COMPACT STATION               |
|                                     |
| +-----+ +-----+ +-----+ +-----+ |
| | PVS980-58 INVERTER | | MV TRANSFORMER | | SF6 SWITCHGEAR | |
| | - 1500V DC Input   | | - 600V / 33 kV  | | - 630A VCB      | |
| | - Thermosiphon Cool |---->| - ONAN Oil Immersed |---->| - REJ603 Relays |==> 33 kV Line
| | - 4,348 kVA @ 25°C | | - 3,000 kVA Continuous| | - 25 kA / 3 sec | |
| +-----+ +-----+ +-----+ +-----+ |
+-----+

```

Thermosiphon Closed-Loop Cooling Physics

Unlike traditional forced-air or liquid-pumped central inverters, the PVS980-58 employs a phase-change **thermosiphon cooling technology**:

- **Phase-Change Cycle:** A non-conductive dielectric liquid refrigerant absorbs heat directly from the insulated gate bipolar transistor (IGBT) power modules, vaporizing at low temperatures.

- **Passive Circulation:** The vapor rises naturally to an external air-cooled condenser heat exchanger mounted on the outer enclosure wall. Ambient air cools the condenser fins, causing the vapor to condense back into a liquid state and drain by gravity back to the IGBT cold plate.
- **Operational Reliability:** This self-contained phase-change loop operates without mechanical liquid pumps, valves, or active plumbing, eliminating coolant leak vulnerabilities and lowering auxiliary parasitic power consumption to  $\leq 2,500$  W during full-load operation.

AC Output Thermal Derating Profile

| AMBIENT TEMPERATURE | COMPACT STATION AC RATING | 3.0 MVA GRID EXPORT MARGIN          |
|---------------------|---------------------------|-------------------------------------|
| At 25°C Ambient     | 4,348 kVA                 | +1,348 kVA Headroom (Zero Clipping) |
| At 35°C Ambient     | 4,229 kVA                 | +1,229 kVA Headroom (Zero Clipping) |
| At 50°C Peak Summer | 3,845 kVA                 | +845 kVA Headroom (Zero Clipping)   |

1.4 Step-Up Transformer & 33 kV Switchgear Specifications

- **Transformer Type:** ONAN (Oil Natural Air Natural) mineral-oil-immersed outdoor transformer.
- **Voltage Ratio:** 0.615 kV low-voltage inverter output stepped up to 33 kV high-voltage grid export.
- **Vector Group & Winding:** Dyn11 vector configuration, with primary delta winding suppressing 3rd-order harmonic currents from entering the utility grid. Winding material is high-grade electrolytic copper/aluminum with an off-load tap changer ( $\pm 2 \times 2.5\%$ ).
- **Medium Voltage Switchgear:** Compact SF6-insulated metal-enclosed switchgear panel housing a 630 A, 25 kA (for 3 seconds) vacuum circuit breaker (VCB).
- **Protection Relay Scheme:** Microprocessor-based ABB REJ603 protection relay providing instantaneous overcurrent (ANSI 50), time-delay phase overcurrent (ANSI 51), earth fault (ANSI 50N/51N), and transformer Buchholz/temperature trip outputs.

02 Civil, Structural & Meteorological Mechanics

2.1 Geotechnical Soil Analysis & Foundation Engineering

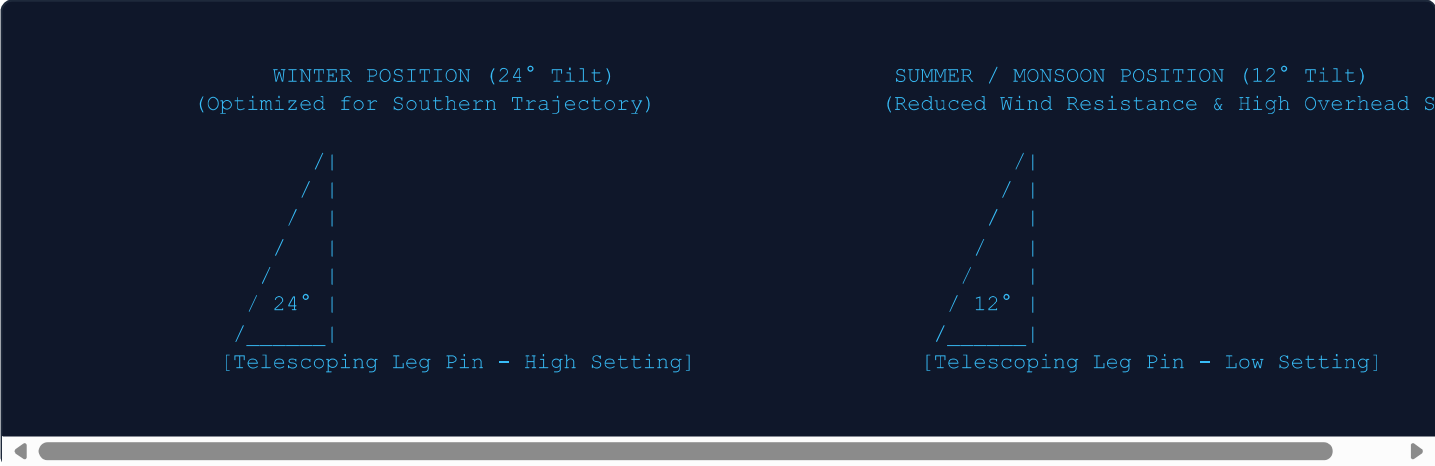
The project site in Santuri, Purulia district features typical Chota Nagpur plateau geology, characterized by hard gravelly red soil underlaid by weathered granite-gneiss rock at shallow depths (1.2 m to 2.0 m).

```
Surface Level (0.0 m)
===== <-- Grade Level
| Gravelly Red Soil / Sandy Clay |
| (Allowable Bearing Capacity: ~120-150 kN/m²) |
|----- <-- 1.0 m Depth
| Weathered Granite-Gneiss Bedrock |
| (High Passive Resistance & Lateral Stability) |
|----- <-- Pile Tip (1.5 m Depth)
| Reinforced Concrete Bored Pile (M30 Grade) |
| Diameter: 300 mm | Galvanized Steel Post Anchored |
=====
```

- **Pile Foundation Specification:** Reinforced concrete bored cast-in-situ piles using M30 grade concrete with sulphate-resistant cement.
- **Pile Dimensions:** Diameter 300 mm, driven to a minimum depth of 1.5 meters into the native soil/rock matrix.
- **Pull-Out & Uplift Resistance:** Designed to resist tensile uplift forces generated by high monsoon wind suction. Each pile foundation undergoes mandatory pull-out testing to withstand a minimum vertical uplift force of 18 kN and a lateral shear force of 12 kN.

## 2.2 Structural Load Dynamics & Seasonal Tilt Mechanics

The mounting structures utilize **Hot-Dip Galvanized Iron (HDG)** structural steel sections with a zinc coating thickness of  $\geq 80$  microns ( $\sim 570 \text{ g/m}^2$ ), providing a corrosion resistance lifespan exceeding 25 years in atmospheric corrosion category C3. Certified via STAAD.Pro structural analysis to withstand basic wind speeds up to **170 km/h** (IS 875 Part 3: 2015).



**Seasonal Yield Advantage:** Shifting tables from 12° (Summer/Monsoon) to 24° (Winter) yields an additional **3% to 5% annual energy generation** (+135,000 kWh to +225,000 kWh per year) compared to a single fixed 23° installation, requiring only 3 days of low-cost manual labor twice a year.

## 2.3 Water Hydrology, Demineralization & Soiling Management

Dust accumulation (soiling) on glass surfaces reduces light transmittance, causing an unmanaged generation loss of up to 0.5% per day during dry months in Purulia.

|                                 |                          |                             |                        |
|---------------------------------|--------------------------|-----------------------------|------------------------|
| +-----+ +-----+ +-----+ +-----+ |                          |                             |                        |
| Deep Borewell                   | ====>  Dual-Media Filter | ====>  Reverse Osmosis (RO) | ====>  Demineralized W |
| Raw Water Source                | Suspended Solids         | Desalination Unit           | Storage Tank           |
| Total Dissolved                 | Removal                  | TDS Reduction               | TDS < 50 ppm           |
| Solids > 450 ppm                | +-----+                  | < 50 ppm Target             | pH 6.5 - 7.5           |
| +-----+                         |                          | +-----+                     | +-----+                |

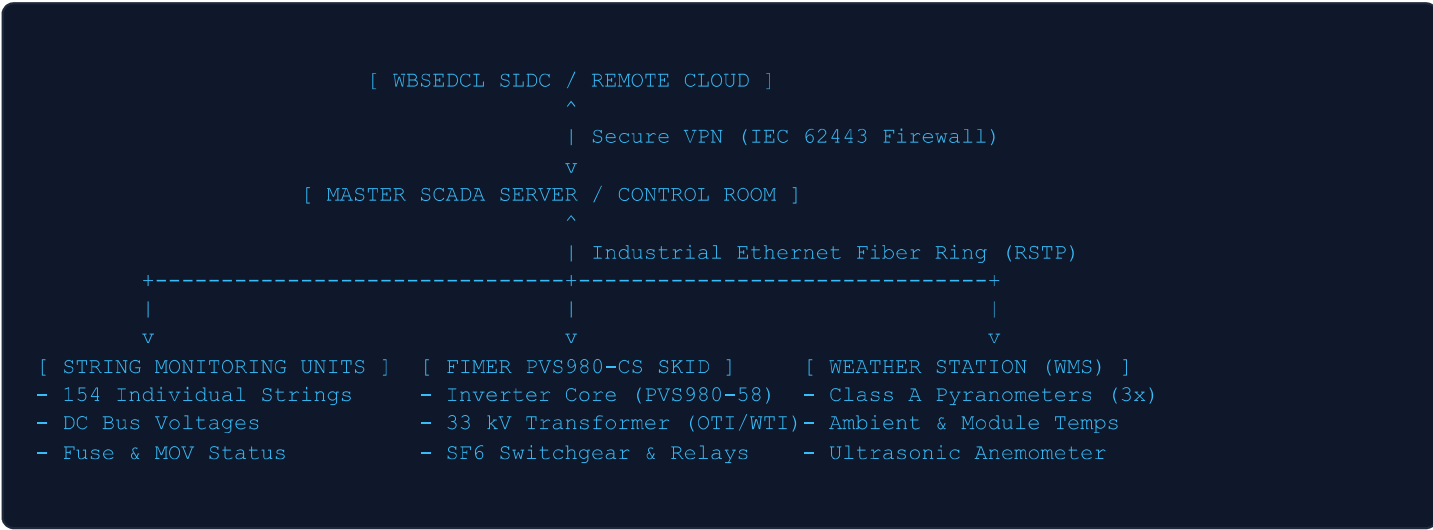
- **On-Site RO Plant:** On-site 150 mm deep borewell (5 HP pump, 5,000 L/hr) paired with a Dual-Media Filter and Reverse Osmosis plant to reduce TDS from ~450 ppm to **< 50 ppm** (pH 6.5–7.5), preventing calcification and permanent glass scaling.
- **Cleaning Schedule:** 24 cleaning cycles per year (bi-monthly during non-monsoon months). Each cycle consumes 2.5 liters of DM water per module (10,780 liters per full plant wash) at 20 bar pressure.

03

## Monitoring, Control & Data Acquisition Infrastructure

### 3.1 SCADA System Architecture & Network Topology

The plant employs a **Master Supervisory Control and Data Acquisition (SCADA)** system built on industrial PLC/IPC hardware running IEC 60870-5-104 and Modbus TCP/IP protocols.



- **Redundant Fiber Backbone:** Single-Mode Fiber Optic (SMFO) ring running Rapid Spanning Tree Protocol (RSTP) ensures < 20 ms communication failover and complete EMI immunity across the 11-acre site.
- **Master Industrial PC:** Stationed in the site control room with RAID-1 SSD storage for 5 years of local logs and a 2-hour UPS battery backup.

### 3.2 String Monitoring Units (SMU) & Array Diagnostics

Data is collected at the **String Monitoring Unit (SMU)** level across all **154 parallel strings** using Hall Effect current sensors ( $\pm 0.5\%$  accuracy).

```
String 1 (28 Modules) ---> [ Hall Sensor 1 ] --+
String 2 (28 Modules) ---> [ Hall Sensor 2 ] --|
...                                     |--> [ Local Microprocessor ] --> Modbus RS485 --> Fiber
String N (28 Modules) ---> [ Hall Sensor N ] --+      - Voltage Sensor
                                                    - Temp Sensor
                                                    - MOV Surge Monitor
```

- **Blown Fuse Alert:** Triggers instantly if string current drops to 0 A while parallel strings draw > 10 A.
- **Soiling / Shading Alarm:** Triggers if a string's output deviates by > 8% from neighboring strings under uniform irradiance.
- **PID / Degradation Tracking:** Tracks slow-onset current loss across strings over consecutive months.

### 3.3 Meteorological Station & Performance Accounting

| SENSOR / METER          | SPECIFICATION                                   | PRIMARY FUNCTION                                   |
|-------------------------|-------------------------------------------------|----------------------------------------------------|
| POA Pyranometer (Front) | Class A Thermopile (ISO 9060:2018), heated      | Measures Plane of Array irradiance at 12°/24° tilt |
| POA Pyranometer (Rear)  | Class A Thermopile, downward facing             | Measures ground-reflected albedo gain              |
| Module Temp Sensors     | Pt100 4-wire RTD (4 units glued to modules)     | Calculates cell temp and thermal voltage derating  |
| Ultrasonic Anemometer   | Solid-state ultrasonic sensor (no moving parts) | Measures wind speed (0 to 60 m/s) and direction    |
| ABT Tariff Meters       | Dual 0.2s Class Main/Check Tri-Vector Meters    | Revenue accounting & GPRS AMR streaming to SLDC    |

$$PR = [ E\_AC\_Exported (kWh) / P\_DC\_Installed (kWp) ] \div [ H\_POA (kWh/m^2) / G\_STC (1 kW/m^2) ] \times 100\%$$

If the 15-minute rolling Performance Ratio (PR) drops below **75.0%** under unshaded conditions (POA > 600 W/m²), an automated SCADA high-priority alert is raised.

### 3.4 Site Security & Cybersecurity Architecture

- **Perimeter CCTV:** 12–16 HD IP Bullet Cameras with IR night vision (50 m) plus 2 High-Speed 360° PTZ Dome Cameras (30x optical zoom) with AI motion analytics.
- **IEC 62443 Cyber Security:** Industrial Stateful Inspection Firewalls segregate OT networks from IT/Internet traffic. Remote access uses encrypted IPSec/OpenVPN tunnels with MFA. SLDC telemetry uses outbound-only read-only channels.

### 4.1 Capital Expenditure Engineering Breakdown

The project requires a consolidated capital outlay of **₹10.80 Crore**, structured to achieve a debt-to-equity ratio of **70:30**.

| CAPEX COST CENTER  | GRANULAR COMPONENT SCOPE                                      | TOTAL COST   | % SHARE |
|--------------------|---------------------------------------------------------------|--------------|---------|
| PV Modules         | 4,340 Waaree 700W N-type TOPCon Modules (140 pallets)         | ₹5.60 Crore  | 51.9%   |
| Power Conversion   | FIMER PVS980-CS 4.3 MVA Compact Station & Switchgear          | ₹2.00 Crore  | 18.5%   |
| Civil & Structural | Piling, HDG seasonal tilt tables, boundary wall, roads, store | ₹1.10 Crore  | 10.2%   |
| Grid Evacuation    | 3.0 km dedicated 33 kV line, poles, 33 kV bay at WBSEDCL      | ₹0.90 Crore  | 8.3%    |
| Land Procurement   | 11.0 Acres contiguous freehold land & conversion fees         | ₹0.60 Crore  | 5.5%    |
| Soft Costs & Fees  | Engineering, CEIG inspection fees, insurance, contingency     | ₹0.60 Crore  | 5.6%    |
| TOTAL CAPEX        | COMPLETE TURNKEY PROJECT INFRASTRUCTURE                       | ₹10.80 CRORE | 100.0%  |

### 4.2 Capital Financing Structure & Bank Loan Amortization

- **Promoter Equity Contribution (30%):** ₹3.23 Crore
- **Green Energy Bank Loan (70%):** ₹7.57 Crore at 8.50% p.a. over a **10-Year Base Sanction Tenure**.
- **Scheduled Annual Bank EMI:** ₹1.13 Crore per year (Monthly EMI: ₹9.39 Lakhs), delivering a bank-compliant **1.50x Scheduled DSCR** in Year 1.
- **Accelerated Debt Payoff Strategy:** Director salary is set to **₹0.00 Cr** during the debt phase, and a permanent **₹10.00 Lakhs working capital float** is retained in Year 1. 100% of remaining operational cash flow is applied as principal prepayments, **fully extinguishing the bank loan in 5.3 years (Month 5 of Year 6)** and saving **₹1.69 Crore** in interest.

### 4.3 Income Tax Code Optimization (Section 32 & Section 115BAA)

```
Gross Revenue (Yr 1): ₹1.83 Cr
|
v
(-) Plant OpEx: -₹0.10 Cr ==> EBITDA: ₹1.73 Cr
|
v
(-) Bank Interest (Sec 36): -₹0.62 Cr
|
v
(-) Accelerated Depreciation (Sec 32 - 40% WDV): -₹3.52 Cr
|
v
Taxable Corporate Income: -₹2.41 Cr (Net Operating Loss Carried Forward)
|
v
CORPORATE TAX PAYABLE (Years 1 to 6): ₹0.00
```

- **Accelerated Depreciation (Section 32):** Claiming 40% WDV accelerated depreciation in Year 1 (~₹3.52 Cr) creates an artificial tax shield, resulting in **₹0 corporate income tax liability for Years 1 through 6**.
- **Concessional Tax Rate (Section 115BAA):** Opting into Section 115BAA sets an effective flat corporate tax rate of **25.17%** (22% base + 10% surcharge + 4% cess) with complete Minimum Alternate Tax (MAT) exemption.

### 4.4 Corporate Treasury Wealth Compounding & Director Salary Model

Once the bank debt is fully cleared in Month 5 of Year 6, the company functions as an unencumbered cash generator. The company adopts a structured dual-cash allocation strategy over the remaining 19.5 compounding years (Years 6 to 25):

*Annual Post-Tax Net Cash Surplus (~₹1.05 Cr to ₹1.18 Cr) ==> Fixed ₹1.00 Cr / Yr to Mutual Funds + Residual Cash Drawn as Director Salary*

- **Fixed Corporate Mutual Fund Allocation:** The company deploys a fixed **₹1.00 Crore per year** into a high-yield corporate mutual fund portfolio generating an assumed **12.0% CAGR** over 19.5 years.  
*Total Principal Invested: 19.5 years × ₹1.00 Cr = ₹19.50 Crore.*  
*Gross Portfolio Value at Year 25: ~₹72.50 Crore.*
- **Director Compensation (Residual Draw):** All extra post-tax operational cash flow remaining after funding the ₹1.00 Crore annual mutual fund investment (~₹0.05 Cr to ₹0.06 Cr/yr in early post-debt years) is drawn directly by the director as variable compensation (~₹0.80 Cr cumulative total over plant lifecycle).

## 4.5 Terminal Company Liquidation Waterfall (Year 26)

```
[ TOTAL COMPANY GROSS ASSETS: ~₹76.00 Cr ]
- Mutual Fund Portfolio: ₹72.50 Cr
- Scrap Steel, Copper & 11 Acres Land: ₹3.50 Cr
  |
  |---> (-) Corporate LTCG Tax on MFs (12.5% under Sec 112): -₹6.62 Cr
  |---> (-) Corporate Tax on Asset Scrap (25.17%): -₹0.90 Cr
  |
  v
[ NET DISTRIBUTABLE COMPANY CASH BALANCE: ₹68.48 Cr to ₹69.50 Cr ]
  |
  +-----+
  |
  |
  v
[ OPTION A: Single-Year Exit ]      [ OPTION B: 5-Year Staggered Exit ]
- Dividend Tax (Sec 2(22)(e)): ~38%  - Capital Reduction / Buyback Spread
- Tax Outflow: -₹25.48 Cr to -₹26.50 Cr - Tax Outflow: -₹20.28 Cr to -₹21.30 Cr
- NET IN-HAND: ~₹43.00 Crore         - NET IN-HAND: ~₹48.20 Crore (+ ₹0.80 Cr Salary)
```

## 05 Regulatory Code, Statutory Permitting & Grid Interconnection

### 5.1 WBLR Act Section 4C Land Conversion

Developing a 3.02 MWp plant on 11.0 acres in Santuri requires formal land conversion under **Section 4C of the West Bengal Land Reforms Act, 1955** from *Agricultural* to *Industrial - Solar Power Generation*.

```
[ Land Title Acquisition ] ==> [ Application to DL&LRO Purulia ] ==> [ BL&LRO Site Inspection ]
- 11.0 Acres Freehold          - Form 1 under Sec 4C              - Verify Non-Irrigated
- Deed Registration            - Mutation Certificates          - Confirm Single Crop
                                   |
                                   v
[ Conversion Clearance Issued ] <== [ Clearance from WB Land Board ] <== [ Payment of Conversion Fee ]
- Issued within 60 Days          DL&LRO Clearance              - Rs 15,000 / Acre
```

### 5.2 Grid Interconnection Standards & WBSEDCL 2025 Code

- **Harmonic Limits:** Central inverter maintains Total Current Harmonic Distortion (THDi) < 3.0%, complying with CEA Technical Connectivity Standards.
- **Anti-Islanding Protection:** Disconnects from the grid in < 100 ms upon utility outage to protect line maintenance crews.
- **Ride-Through Capability:** Features Low Voltage Ride Through (LVRT) down to 15% voltage for 150 ms and operates across 47.5 Hz to 52.0 Hz grid frequency limits.

### 5.3 Statutory Electrical Safety Mandate (CEIG Clearance)

**Mandatory CEIG Inspection:** Under Regulation 43 of CEA Safety Regulations 2010, installations above 11 kV cannot be self-certified. A physical site audit by the **Chief Electrical Inspector to the Government of West Bengal (CEIG)** is legally mandatory prior to energization.

Under WBSEDCL Procedures-C (2025), plants exceeding 170 kW capacity must undergo mandatory safety re-inspection and submit compliance certificates **every two years** throughout their 25-year operational lifecycle.

## 06 Consolidated Project Life-Cycle Comparison

| PARAMETER                 | INITIAL BASELINE MODEL              | FULLY OPTIMIZED UTILITY MODEL            | ENGINEERING ADVANTAGE                           |
|---------------------------|-------------------------------------|------------------------------------------|-------------------------------------------------|
| DC Array Capacity         | 3.00 MWp (4,290 Modules)            | 3.02 MWp (4,340 Modules)                 | +18.4 kWp higher generation                     |
| String Sizing             | 143 Strings × 30 Modules            | 154 Strings × 28 Modules                 | Eliminates 1,500V cold-winter overvoltage       |
| Power Conversion          | 2.5 MVA Inverter Skid               | FIMER PVS980-CS 4.3 MVA Skid             | 3.845 MVA AC rating even at 50°C summer         |
| Grid Evacuation           | 2.50 MVA Interconnection            | 3.00 MVA Dedicated Feeder                | Zero AC thermal clipping headroom               |
| Loan Structure & DSCR     | 7-Year Base / 1.07x DSCR            | 10-Yr Base (5.3-Yr Payoff) / 1.50x DSCR  | Bank-compliant DSCR + saves ₹1.69 Cr interest   |
| Capital Expenditure       | ₹10.10 Crore                        | ₹10.80 Crore                             | Includes full 3 MVA line & 4,340 modules        |
| Financing Split           | ₹3.03 Cr Equity / ₹7.07 Cr Loan     | ₹3.23 Cr Equity / ₹7.57 Cr Loan          | Maintains optimal 70:30 debt leverage           |
| Post-Debt Cash Allocation | Fixed ₹1.05 Cr MF / ₹0.25 Cr Salary | Fixed ₹1.00 Cr MF / Residual to Director | Guarantees ₹1.00 Cr compounding + flexible draw |
| Year 25 Treasury Corpus   | ₹54.00 Crore                        | ₹72.50 Crore                             | +₹18.50 Cr higher mutual fund accumulation      |
| YEAR 26 NET HARVEST       | ₹31.30 CRORE                        | ~₹43.00 CR TO ₹48.20 CR                  | UP TO +₹16.90 CR HIGHER NET LIQUID WEALTH       |